

Null Evidence Audit: Verifying the Absence of Human Organ Regeneration Data in Non-Biomedical AI Literature

Input Question

Can a human tissue or organ be fully regenerated?

Domain

Medicine & Health

Validation Status

needs_data

Problem Statement

The question asks whether full regeneration of human tissues or organs is currently achievable, and what biological mechanisms or technological pathways (in vitro vs. in vivo) are involved. The scope is limited to verifying this capability against the strictly bounded set of allowed evidence IDs, which are known to be non-biomedical.

Rationale

Scientific literature often discusses the potential of regenerative medicine, drawing parallels with species like salamanders. However, to establish a factual claim within the SAGE125 framework, one must rely exclusively on provided EvidenceCards. The current allowed evidence set consists of papers on AI philosophy, data ethics, and image retrieval. Therefore, the primary scientific task is to rigorously verify the absence of relevant biomedical data in this specific corpus, framing it as a testable null hypothesis rather than making unsupported biological claims.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Within the strictly bounded corpus of allowed evidence IDs, there exists no verified factual assertion confirming that human tissues or organs can be fully regenerated in vivo or in vitro.
- **Mechanism**: The allowed evidence set (EV-Q016-cca15123cafdcd183383a782, EV-Q016-4e2c3db0dc947ff33a07e437, EV-Q016-bcd94dc9b73d0cdfed28d5fc, EV-Q016-aa66383ea6a3ea35fb257f48) is semantically restricted to AI philosophy, data ethics, and image retrieval methodologies. Consequently, any query regarding biological regeneration mechanisms or clinical efficacy must return a null result when validated against this specific knowledge base.
- **Falsifiable Prediction**: A deterministic audit using pre-specified keyword-context rules will identify exactly zero sentences in the allowed evidence IDs that assert successful full human organ regeneration as a verified fact.

- **Required Observations**: Complete text extraction from all four allowed Evidence IDs. ; Application of a deterministic rule set (not a black-box model) to classify content as 'Biomedical Regeneration Fact' vs 'Non-Biomedical'. ; Verification that booklet excerpts are excluded from the factual verification process.
- **Risk of Being Wrong**: Low risk under current constraints; however, this hypothesis is conditionally valid only as long as no biomedical evidence is added to the allowed_evidence_ids list. Addition of relevant biological studies would immediately falsify this null finding.

Hypothesis 2

- **Hypothesis**: General methodological definitions of 'problem-solving activity' found in non-biomedical literature can serve as a reproducible structural template for defining regenerative medicine research gaps, independent of domain-specific biological data.
- **Mechanism**: Abstract definitions of methodology and training (EV-Q016-aa66383ea6a3ea35fb257f48) provide domain-agnostic logical scaffolding. When applied to the 'knowledge gap' state identified in the evidence extraction, these definitions structure the inquiry into verifiable sub-questions rather than unstructured speculation.
- **Falsifiable Prediction**: Mapping the definitions from EV-Q016-aa66383ea6a3ea35fb257f48 to the regeneration question yields a structured research framework with distinct, actionable components, whereas random text selection from the same source fails to produce such structure.
- **Required Observations**: Extraction of 'Methodology' and 'Training' definitions from EV-Q016-aa66383ea6a3ea35fb257f48. ; Demonstration of a logical mapping between these abstract definitions and specific regenerative medicine unknowns. ; Comparative analysis showing superior structure compared to a baseline of unrelated text segments.
- **Risk of Being Wrong**: High risk; the semantic distance between AI image retrieval methodology and wet-lab regenerative medicine is significant. The apparent structural utility may be coincidental or overly generic to be scientifically meaningful.

Technical Details

This experiment validates the null hypothesis that the strictly bounded corpus of allowed Evidence IDs (EV-Q016-cca15123cafdcd183383a782, EV-Q016-4e2c3db0dc947ff33a07e437, EV-Q016-bcd94dc9b73d0cdfed28d5fc, EV-Q016-aa66383ea6a3ea35fb257f48) contains no verified factual assertions regarding full human organ regeneration. To address reviewer feedback on reproducibility, the methodology replaces black-box NLP models with a deterministic, rule-based content audit. The process involves: (1) Text extraction from the four specified arXiv PDFs; (2) Application of a pre-specified keyword-context rule set to identify sentences containing biomedical regeneration terms (e.g., 'human organ', 'tissue regeneration', 'in vivo repair'); (3) Contextual exclusion filtering to remove metaphorical uses of 'regeneration' common in AI/ethics literature (e.g., 'regenerating trust', 'model regeneration'); (4) Final verification that zero sentences meet the criteria for 'Verified Biomedical Fact'. This approach ensures that the absence of evidence is demonstrated through transparent, reproducible logic rather than opaque model predictions.

Datasets

Source

```
```json
[
 {
 "id": "EV-Q016-cca15123cafdcd183383a782",
 "type": "text",
 "description": "ArXiv paper on AI philosophy and cognitive science limits."
 },
 {
 "id": "EV-Q016-4e2c3db0dc947ff33a07e437",
 "type": "text",
 "description": "ArXiv paper on HALE data principles and EU Charter compliance."
 },
 {
 "id": "EV-Q016-bcd94dc9b73d0cdfed28d5fc",
 "type": "text",
 "description": "ArXiv paper on Human-AI collaborative image retrieval explanations."
 },
 {
 "id": "EV-Q016-aa66383ea6a3ea35fb257f48",
 "type": "text",
 "description": "ArXiv paper defining general methodology and training concepts."
 }
]
```
```

Target

```
```json
{
 "type": "structured_audit_log",

```

"description": "A binary classification result for each evidence ID indicating presence (1) or absence (0) of verified human organ regeneration data, along with a log of all keyword hits and their contextual rejection reasons."

}

...

## Paper Abstract

Background: Regenerative medicine aims to replace dysfunctional human body parts, potentially revolutionizing treatment for trauma and disease. While animal models like salamanders demonstrate natural regeneration, the feasibility of full human organ regeneration remains a critical scientific question. Methods: This study conducts a rigorous audit of four specific arXiv papers (EV-Q016-cca15123cafdcd183383a782, EV-Q016-4e2c3db0dc947ff33a07e437, EV-Q016-bcd94dc9b73d0cdfed28d5fc, EV-Q016-aa66383ea6a3ea35fb257f48) to determine if they contain verified facts supporting human organ regeneration. We employ a deterministic, rule-based keyword and context analysis to distinguish between biomedical facts and metaphorical uses of 'regeneration' in AI and ethics literature. Validation Plan: The protocol involves text extraction, strict keyword matching for biomedical terms, and contextual exclusion of AI-related metaphors. Results: pending (待执行验证实验). The study expects to confirm the absence of relevant biomedical data in this specific corpus, highlighting the importance of evidence grounding in scientific inquiry.

## Methods

1. Deterministic Text Extraction: Parse full text from the four allowed Evidence IDs. 2. Rule-Based Keyword Scanning: Apply strict regular expressions for terms like 'human organ regeneration', 'full tissue regrowth'. 3. Contextual Exclusion Rules: Filter out matches associated with AI/ethics metaphors (e.g., 'algorithm', 'data subject'). 4. Null Hypothesis Verification: Count remaining sentences; support hypothesis if count is zero.

## Experiments

### Baselines

```json

[

"Random Keyword Match: Counts any occurrence of the word 'regeneration' regardless of context, serving as a baseline to demonstrate the high rate of false positives from metaphorical usage in AI literature.",

"Broad Medical Dictionary Filter: Uses a general medical term list to identify any health-related mentions, demonstrating the specificity gap between general health terms and the specific claim of 'full human organ regeneration'."

]

...

Metrics

```json

[

"Precision of Biomedical Claims: Proportion of identified 'regeneration' mentions that are actually about human organs (Expected: 0% given the corpus composition).",

"Evidence Coverage Score: Percentage of allowed Evidence IDs containing at least one verified fact about human organ regeneration (Expected: 0%).",

"False Positive Rate: Rate at which AI/ethics metaphors are incorrectly classified as biomedical facts by the baseline keyword matcher, highlighting the necessity of the contextual exclusion rules."

]

'''

## Ablation

Remove the contextual exclusion rules to rely solely on keyword matching, demonstrating the high risk of misinterpretation without deterministic context-aware analysis.

## Validation Protocol

Manual review of all flagged segments by a domain expert (simulated via strict rule-based checking against known AI ethics terminology from the source texts) to confirm that no valid biomedical data exists in the corpus. All decision rules are logged for auditability.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- \*\*EV-Q016-cca15123cafdcd183383a782\*\* · arxiv · arXiv:2202.04977
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2202.04977.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=fb996847c2cd2dc76e59740e3c05847167990c0d4365487732616494f59689e5
- \*\*EV-Q016-4e2c3db0dc947ff33a07e437\*\* · arxiv · arXiv:2209.09724
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2209.09724.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:2|section:page-2|paragraph:1;  
content\_sha256=f39426f5e8aa1d9df0749f431025245098cd06874e9d8746725045f9fcae3471

- **\*\*EV-Q016-bcd94dc9b73d0cdfed28d5fc\*\*** · arxiv · arXiv:1904.03285
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1904.03285.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=5eee51a71908eae7a3d7b32dc73b0ae15116eb72c3363653e09d300b646bca01
- **\*\*EV-Q016-aa66383ea6a3ea35fb257f48\*\*** · arxiv · arXiv:2504.14689
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2504.14689.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:4|section:page-4|paragraph:1;  
content\_sha256=69fd5e63df12b394264ad43a97c4b1a7f9ed352bd27fdae559fc16fc0d1fc3

## Reviewer Comments

- The candidate hypothesis correctly maintains the null state regarding human organ regeneration, strictly adhering to the constraint that allowed Evidence IDs are non-biomedical.
- Experiment design successfully addresses previous reproducibility concerns by replacing the black-box NLP model with a deterministic, rule-based keyword and context audit protocol.
- Results field is correctly marked as pending with no fabrication of experimental outcomes.
- All dataset sources map directly to valid allowed\_evidence\_ids; no external or fabricated references are introduced.
- Baselines (Random Keyword Match) and metrics (Precision of Biomedical Claims) are appropriate for validating the absence of evidence in a text corpus.

## Revision History

- auto\_revision\_1: 依据评审意见重做假设与实验设计。

## Reproducibility Checklist

- Verify access to all four allowed Evidence IDs via their provided URLs.
- Ensure the keyword and exclusion rules are explicitly defined in code and not derived from hidden training data.
- Document all keyword hits and their contextual classification/rejection reasons for transparency.
- Confirm that no external biomedical databases (e.g., PubMed, ClinicalTrials.gov) are queried, strictly limiting scope to allowed IDs.
- Validate that the deterministic rule set produces identical results across multiple runs.